WTF : A (Causal) relationship btw x_1 and y.

A Quick Review of OLS

A simple reg model w/ one exp variable (x_1).

$$y_i = \beta_0 + \beta_1 x_{1i} + u_i \qquad i = \text{obs} \qquad \text{assuming } E(u_i|x_i) = 0$$

ⓐ

$$y_i = \beta_0 + \beta_1 x_{1i} + u_i$$
$$-)\ \bar{y} = \beta_0 + \beta_1 \bar{x}_1 + \bar{u}$$
$$y_i - \bar{y} = \beta_1 (x_{1i} - \bar{x}_1) + (u_i - \bar{u})$$

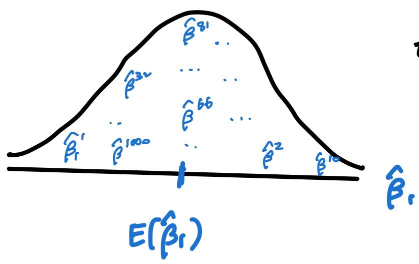
$$\hat{\beta}_1 = \frac{\Sigma(x_{1i} - \bar{x}_1)(y_i - \bar{y})}{\Sigma(x_{1i} - \bar{x}_1)^2} = \frac{cov(x_1, y)}{var(x_1)}$$

What does "bias" mean?

From random sampling with a large sample,

if we estimate β_1 multiple times, we have $\hat{\beta}_1^{1}, \hat{\beta}_1^{2}, \hat{\beta}_1^{3} \ldots \hat{\beta}_1^{1000}$.
(1,000 times)

If we plot all these values and if it looks like this :



then $E(\hat{\beta}_1) = \beta_1$.

On average, estimated $\hat{\beta}_1$ equals the true parameter β_1.

In this case, we say that $\hat{\beta}_1$ is unbiased.

To sum up, we say that $\hat{\beta}_1$ is unbiased if $E(\hat{\beta}_1) = \beta_1$.

Is $\hat{\beta}_1$ from the OLS unbiased?

$$\hat{\beta}_1 = \frac{\Sigma(x_{1i} - \bar{x}_1)(y_i - \bar{y})}{\Sigma(x_{1i} - \bar{x}_1)^2} \overset{ⓐ}{=} \frac{\Sigma(x_{1i} - \bar{x}_1)\left[\beta_1(x_{1i} - \bar{x}_1) + (u_i - \bar{u})\right]}{\Sigma(x_{1i} - \bar{x}_1)^2}$$

$$= \beta_1 + \frac{\Sigma(x_{1i} - \bar{x}_1)(u_i - \bar{u})}{\Sigma(x_{1i} - \bar{x}_1)^2}$$

$$E(\hat{\beta}_1) = E\left[\beta_1 + \frac{\Sigma(x_{1i} - \bar{x}_1)(u_i - \bar{u})}{\Sigma(x_{1i} - \bar{x}_1)^2}\right]$$

$$= \beta_1 + \cancel{E\left[\frac{\Sigma(x_{1i} - \bar{x}_1)(u_i - \bar{u})}{\Sigma(x_{1i} - \bar{x}_1)^2}\right]} \rightarrow \phi$$

because $E[u_i|x_i] = 0$

What if $E[u_i|x_i] \neq 0$?

Then, we have something other than β_1.
$\therefore \hat{\beta}_1$ is biased.

What is Omitted Variable Bias?

Suppose the true model is

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$$

where $E[u_i | x_{1i}, x_{2i}] = \phi$

ⓑ

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$$

$$-)\ \bar{y} = \beta_0 + \beta_1 \bar{x}_1 + \beta_2 \bar{x}_2 + \bar{u}$$

$$y_i - \bar{y} = \beta_1 (x_{1i} - \bar{x}_1) + \beta_2 (x_{2i} - \bar{x}_2) + (u_i - \bar{u})$$

If we mistakenly run a regression w/o x_2,

$$y_i = \beta_0 + \beta_1 x_{1i} + u_i \quad \text{, where } u_i = x_{2i} + \varepsilon_i$$

$$\tilde{\beta}_1 = \frac{\Sigma (x_{1i} - \bar{x}_1)(y_i - \bar{y})}{\Sigma (x_{1i} - \bar{x}_1)^2} \overset{ⓑ}{=} \frac{\Sigma (x_{1i} - \bar{x}_1) \left[\beta_1 (x_{1i} - \bar{x}_1) + \beta_2 (x_{2i} - \bar{x}_2) + (u_i - \bar{u}) \right]}{\Sigma (x_{1i} - \bar{x}_1)^2}$$

$$= \beta_1 + \beta_2 \cdot \frac{\Sigma (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)}{\Sigma (x_{1i} - \bar{x}_1)^2} + \frac{\Sigma (x_{1i} - \bar{x}_1)(u_i - \bar{u})}{\Sigma (x_{1i} - \bar{x}_1)^2}$$

$$E[\tilde{\beta}_1] = E\left[\beta_1 + \beta_2 \cdot \frac{\Sigma (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)}{\Sigma (x_{1i} - \bar{x}_1)^2} + \frac{\Sigma (x_{1i} - \bar{x}_1)(u_i - \bar{u})}{\Sigma (x_{1i} - \bar{x}_1)^2} \right]$$

$$= \beta_1 + \beta_2 E\left[\frac{\Sigma (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)}{\Sigma (x_{1i} - \bar{x}_1)^2} \right] + \cancelto{\phi}{E\left[\frac{\Sigma (x_{1i} - \bar{x}_1)(u_i - \bar{u})}{\Sigma (x_{1i} - \bar{x}_1)^2} \right]}$$

$$= \beta_1 + \beta_2 \cdot \frac{cov(x_1, x_2)}{var(x_1)}$$

Isn't this familiar?

- regress y on x

$$y = a_0 + a_1 x + e$$

$$a_1 = \frac{cov(x, y)}{var(x)}$$

- what we have :

$$\frac{cov(x_1, x_2)}{var(x_1)}$$

$\Rightarrow$ It's the coeff on x_1 from regressing x_2 on x_1

$$x_2 = \delta_0 + \delta_1 x_1 + v$$

$$= \beta_1 + \beta_2 \cdot \delta_1$$

Direct rel btw y & x_1

Direct rel tw x_1 & x_2

Direct rel btw y & x_2

$$= \beta_1 + \underbrace{\beta_2 \delta_1}_{\text{Bias}}$$

To sum up,

$$E(\tilde{\beta}_1) = \beta_1 + \beta_2 \cdot \delta_1$$

- Definition of Unbiasedness

if $\hat{\beta}_1$ is unbiased, $E(\hat{\beta}_1) = \beta_1$.

$$E(\tilde{\beta}_1) = \begin{cases} \beta_1 & \text{if } \beta_2\delta_1 = 0 \text{ ("unbiased")} \Leftrightarrow \beta_2 = 0 \text{ } \underline{\text{OR}} \text{ } \delta_1 = 0 \\ \beta_1 + \beta_2\delta_1 & \text{if } \beta_2\delta_1 \neq 0 \text{ ("biased")} \Leftrightarrow \beta_2 \neq 0 \text{ } \underline{\text{AND}} \text{ } \delta_1 \neq 0 \end{cases}$$

($\because$) we call this term as "BIAS." because it biases our estimation.

What are β_2 and δ_1, again?

β_2 : Direct rel btw y & x_2 from $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$

δ_1 : Direct rel btw x_1 & x_2 from $x_{2i} = \delta_0 + \delta_1 x_{1i} + v_i$

Because this bias is caused by omitting x_2, which is related to BOTH y and x_1, we call this as Omitted Variable Bias (OVB).

Main Takeaway

If we run a regression omitting a variable that is BOTH correlated with y and x_1, our estimate has an OVB problem.
(We CANNOT say that x and y have a causal relationship.)

the variable of interest

(Example) LAB4. Question 7

True model : $\ln E = \beta_0 + \beta_1 \text{indegree} + \beta_2 \text{female} + \beta_3 \text{friend4} + \varepsilon$ (7)

What we estimated (in Q6) : $\ln E = \beta_0 + \beta_1 \text{indegree} + \beta_2 \text{female} + u$ (6)

<Auxiliary regression>

$\text{Friend4} = \delta_0 + \delta_1 \text{indegree} + v$

$= \widehat{4.354} + \widehat{0.082}\ \text{indegree}$

$$E(\tilde{\beta}_1) = \beta_1 + \underbrace{\underset{=0.05}{\hat{\beta}_3} \cdot \underset{=0.082}{\hat{\delta}_1}}_{\text{Bias} > 0}$$

Bias > 0 "upward bias"

the TRUE parameter β_1 from the true model (7)

What we estimated from Equation (6)

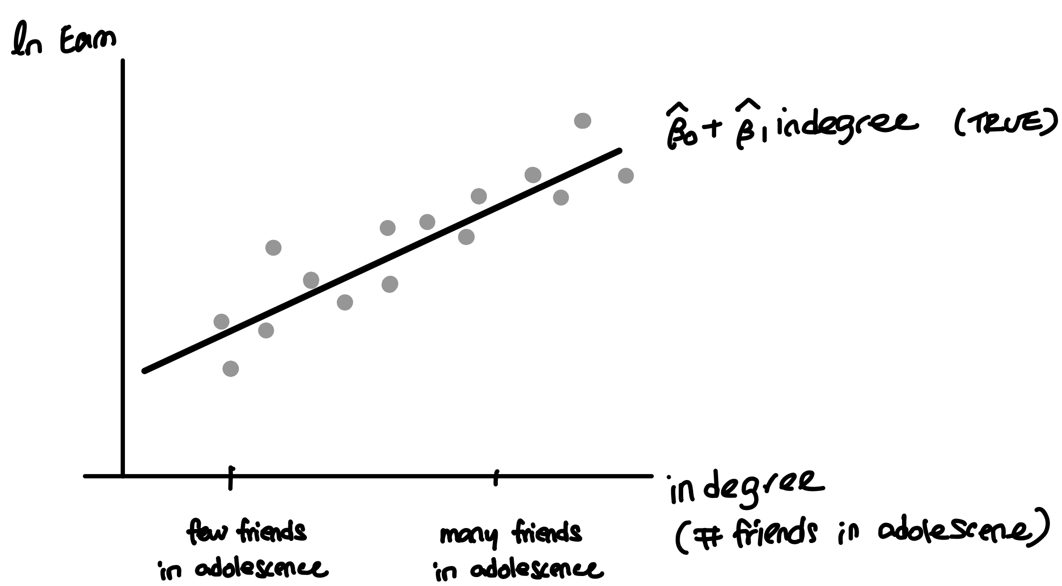
⇒ What can you say about $\tilde{\beta}_1$?

The estimated $\tilde{\beta}_1$ is larger than the true parameter β_1.

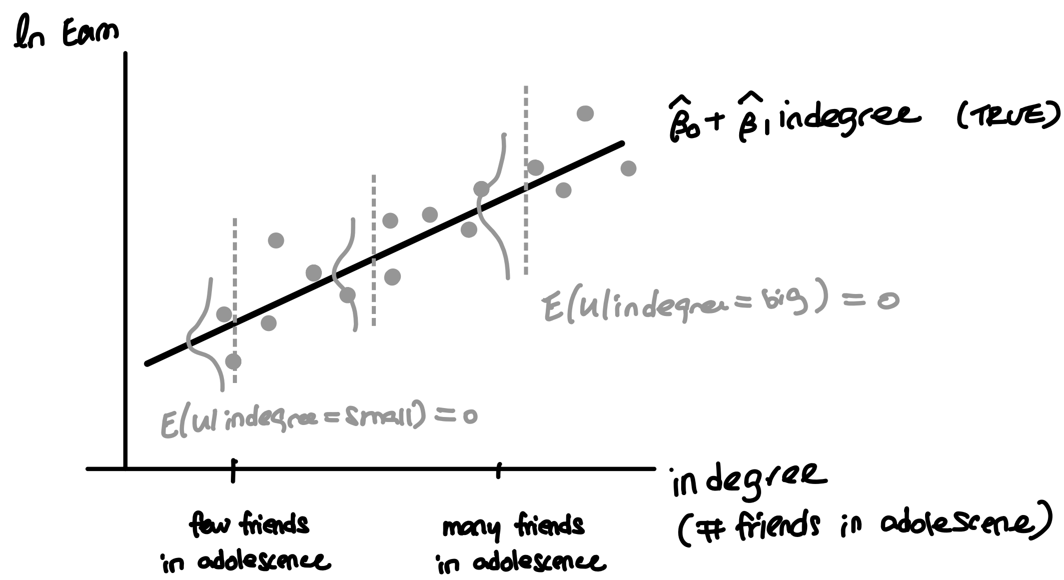
(i.e.) The true parameter is smaller than what we estimated in (6).

We also say that "$\tilde{\beta}_1$ is biased away from zero."

Suppose this is from the true model



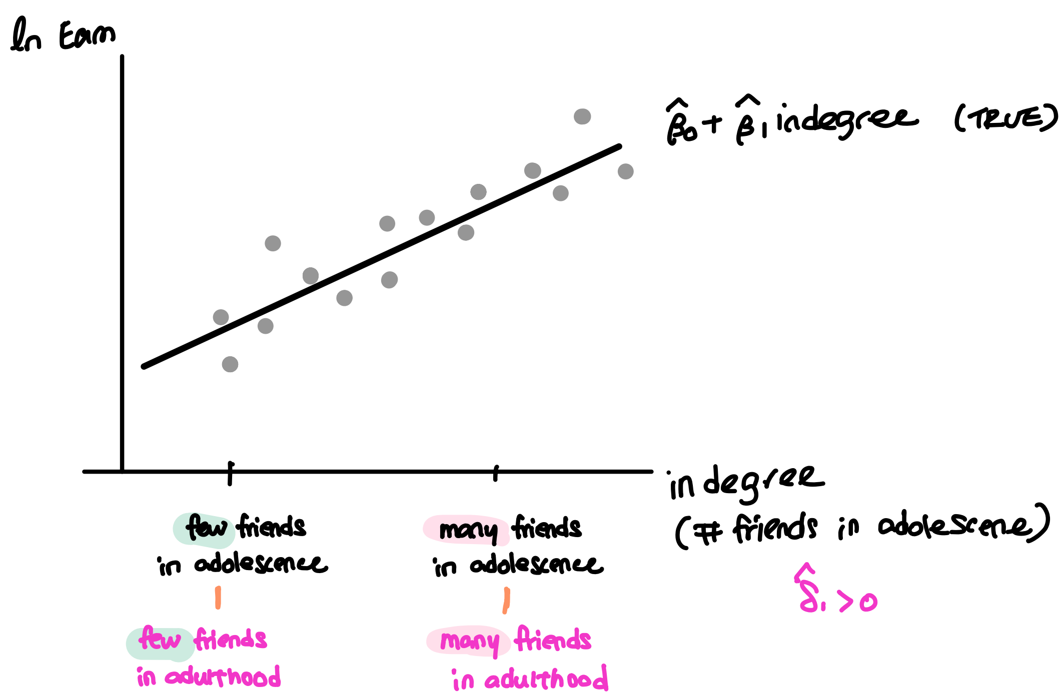
Suppose this is from the true model



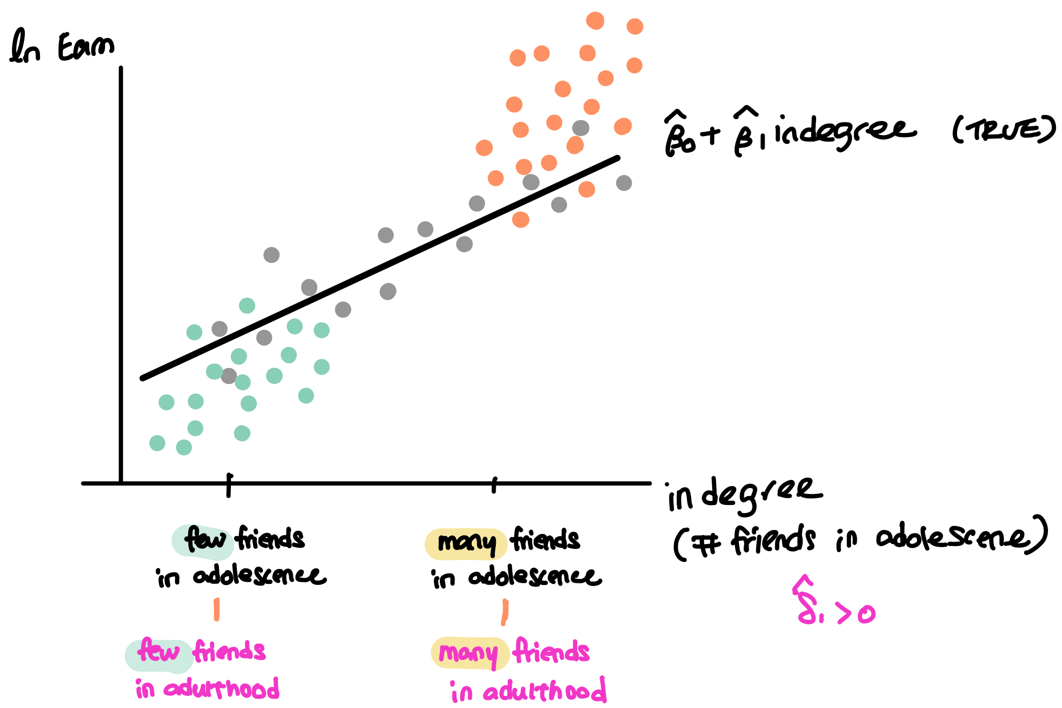
$u = \text{Friend\_4} + \varepsilon$

However, if we omit 'Friend_4' in the model, it is now part of the error term.

$\hat{\delta}_1 > 0$: "A positive rel btw # friends in adolescence & # friends in adulthood."

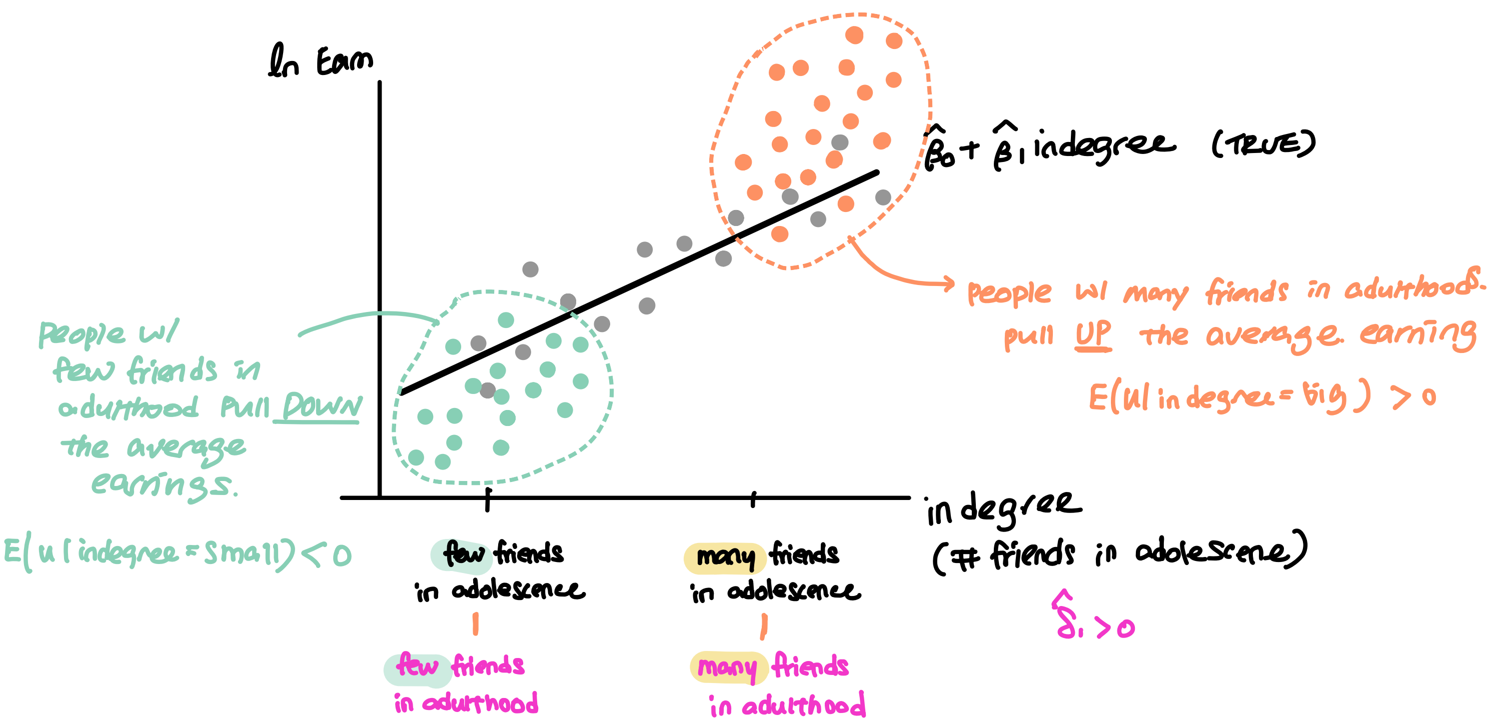


$\hat{\beta}_2 > 0$: "A positive relationship btw Earnings & # friends in adulthood."

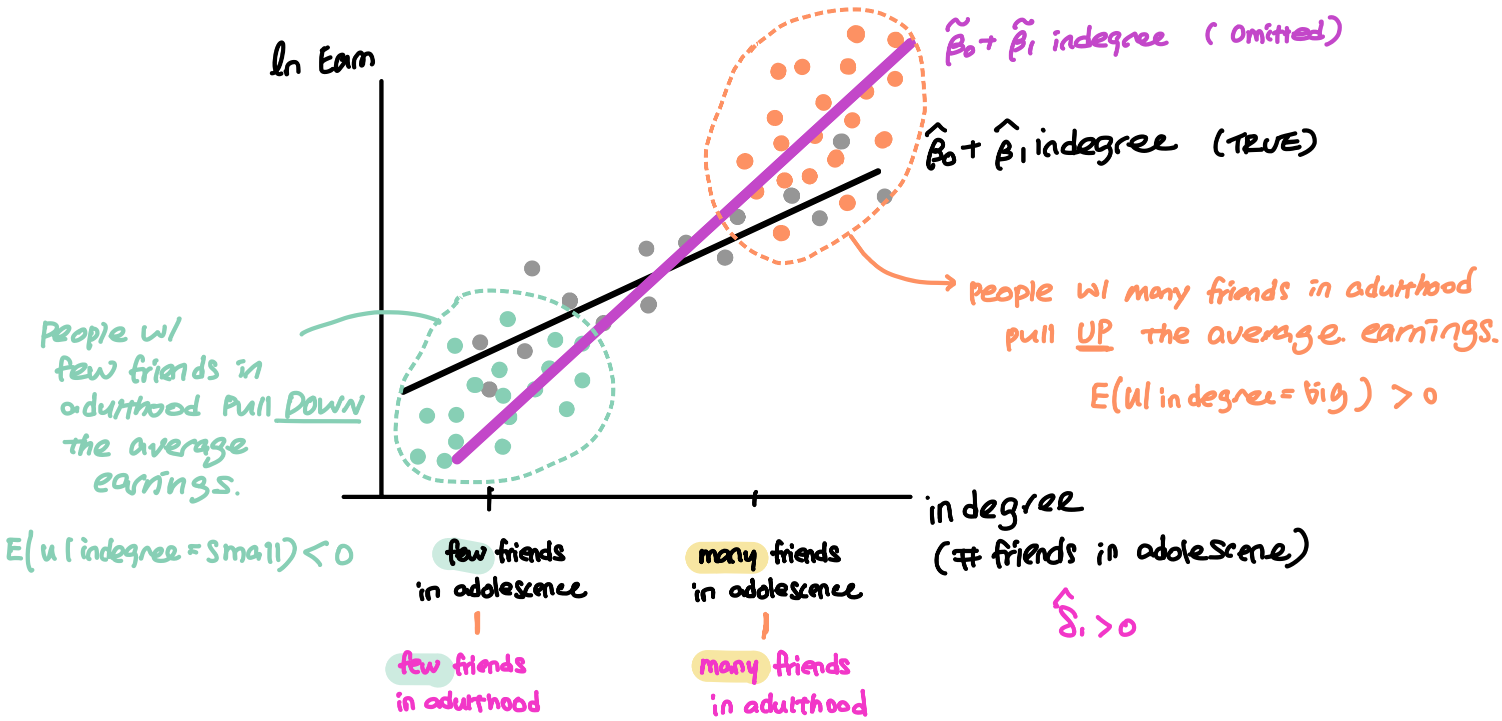


If we omit Friend_4 in the reg model, the error term u = Friend_4 + ε.

Because Friend_4 is correlated with lnEarn AND in_degree, $E[u \mid \text{in\_degree}] \neq \phi$.



Therefore ...



ln Earn
$\tilde{\beta}_0 + \tilde{\beta}_1$ indegree (omitted)
$\hat{\beta}_0 + \hat{\beta}_1$ indegree (TRUE)
People w/ few friends in adulthood pull DOWN the average earnings.
$E(u \mid \text{indegree} = \text{small}) < 0$
people w/ many friends in adulthood pull UP the average earnings.
$E(u \mid \text{indegree} = \text{big}) > 0$
indegree (# friends in adolescence)
few friends in adolescence
few friends in adulthood
many friends in adolescence
many friends in adulthood
$\hat{\delta}_1 > 0$

True model : $\ln Earn = \beta_0 + \beta_1 \, indegree + \beta_2 \, female + \beta_3 \, friend4 + \varepsilon$

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. reg ln_earn in_degree female friend_4
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Source	SS	df	MS
Model	134.501255	3	44.8337517
Residual	2705.73506	2,597	1.04186949
Total	2840.23631	2,600	1.09239858

Number of obs	=	2,601
F(3, 2597)	=	43.03
Prob > F	=	0.0000
R-squared	=	0.0474
Adj R-squared	=	0.0463
Root MSE	=	1.0207

ln_earn	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
$\hat{\beta}_1$ in_degree	.0200568	.005484	3.66	0.000	.0093035	.0308102
female	-.3045655	.0404073	-7.54	0.000	-.3837993	-.2253316
$\hat{\beta}_3$ friend_4	.0503541	.0073086	6.89	0.000	.0360229	.0646853
_cons	10.00184	.0513622	194.73	0.000	9.901127	10.10256

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. reg friend_4 in_degree
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: $Friend4 = \delta_0 + \delta_1 \, indegree + v$ (Auxiliary regression)

Source	SS	df	MS
Model	268.635594	1	268.635594
Residual	22103.3174	2,891	7.64556119
Total	22371.953	2,892	7.7358067

Number of obs	=	2,893
F(1, 2891)	=	35.14
Prob > F	=	0.0000
R-squared	=	0.0120
Adj R-squared	=	0.0117
Root MSE	=	2.7651

friend_4	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
$\hat{\delta}_1$ in_degree	.0824532	.0139101	5.93	0.000	.0551785	.1097279
_cons	4.353976	.0828171	52.57	0.000	4.191589	4.516362